

# RETAIL SALES DATA ANALYSIS

## Data Cleaning & Business Insights Report

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Tools: Microsoft Excel | Power BI

## 1. Executive Summary

This report documents the end-to-end data analysis process applied to a retail sales dataset containing 76,500 rows of transactional data spanning 2022 to 2024. The analysis covered data cleaning in Microsoft Excel and visualization in Power BI.

The dataset represents sales across 10 South African cities, 7 product categories, and includes customer demographics, transaction details, and satisfaction ratings.

### Key Headline Findings

Total Revenue: R448M | Total Transactions: ~75 000 | Avg Order Value: R5.97K | Avg Rating: 3

## 2. Key Findings

### 2.1 Sales Performance

#### Revenue Trend Over Time

Revenue grew steadily from 2022 to 2024, mostly peaking in August and November

### **Best Performing Category**

*Electronics was the top revenue-generating category, accounting for 52.46% of total sales.*

### **Day of Week Patterns**

*Tuesdays recorded the highest transaction volumes, suggesting midweek shopping behavior*

### **Top 10 Products**

*Laptop, Smart TV, and Couch were the top 3 revenue-generating products*

## **2.2 Customer Insights**

### **Age Group Spending**

*The Under 18 age group had the highest average order value at R6.11K, followed by the 18 – 24 group at R6.09K*

### **Gender Breakdown**

*Male and female customers spent comparably, with male customers slightly ahead at R143.01M vs R139.93M*

### **Payment Method Preferences**

*Cash was the most popular payment method at 20.24%, followed by Credit Card at 20.14%*

## **2.3 Regional Performance**

### **Top Performing Region**

*Eastern Cape (Port Elizabeth + East London) generated the most revenue, followed by Gauteng (Pretoria + Johannesburg)*

### **Return Rates by Region**

*Eastern Cape had the highest return rate at 10.11%, which may warrant further investigation*

## **2.4 Returns Analysis**

### **Return Rate by Category**

*Toys had the highest return rate at 14.64%, possibly due to product defects or mismatched expectations*

### 3. Recommendations

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Based on the findings, the following actions are recommended:

- **Customer Targeting:** Focus marketing on the Under 18 age group and top-performing regions where revenue potential is highest
- **Returns Reduction:** Investigate the high return rate in Toys, consider improving product descriptions and after-sales support
- **Promotional Planning:** Schedule promotions around peak transaction days and months to maximize sales volume
- **Payment Access:** Review payment infrastructure in lower-performing regions to ensure all payment methods are available
- **Data Quality:** Standardize data entry at point of sale to reduce cleaning effort in future cycles, particularly for dates and gender fields

### 4. Tools Used

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| Tool             | Purpose                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| Microsoft Excel  | Data cleaning: handling nulls, duplicates, date standardization, casing, validation and formula-based transformation |
| Power BI Desktop | Interactive dashboard: sales overview, customer insights, regional analysis and operational metrics                  |

### 5. Limitations

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- **Synthetic data:** The dataset was artificially generated, findings reflect simulated patterns rather than real business behavior
- **Missing dates:** Approximately 1,041 rows had invalid or missing dates and were excluded from time-series analysis
- **Anonymized customers:** Customer names are anonymized, limiting individual-level analysis

### 6. Conclusion

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This project demonstrated a complete data analysis workflow, from raw, messy data through to actionable business insights. The cleaning process in Excel resolved over a dozen distinct data quality issues, and the resulting dataset enabled meaningful analysis across sales performance, customer behavior, product trends, and regional performance, visualized in an interactive Power BI dashboard.